Withdrawing an Alternative: A Survey of Restricted-Menu Evidence

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Abstract

Told that one of a set of mutually exclusive outcomes has been withdrawn, scratched, delisted or masked out of a softmax, proportional renormalization is the reflex, and in many models the architecture. An analyst holding shares and no restricted-menu observation has two parameter-free options. Renormalization is the Gumbel point of the independent additive random-utility class[2]. Re-running a Gaussian contest among the survivors is Case V. Anything between costs a fitted number. Psychology has tested renormalization under restriction since 1957, under the name constant-ratio rule, and we revisit that question by scoring the two defaults against each other out of sample on thirty-four population comparisons, old and new. The contest wins where alternatives are distinct and unordered, and loses on a perceptual continuum, where a survivor has a near-substitute, or where removal changes the task.

1 A Two Horse Race Between Zero Parameter Choice Models

A population chooses among a menu of alternatives and its shares are known. One or more of the alternatives is then withdrawn: a horse is scratched, a product is delisted, a candidate fails to qualify for a ballot, a response option is disallowed, a token is masked out of a softmax. What are the shares over the survivors?

Nothing has been observed about the smaller menu, so the answer must come from an assumption about how the original shares arose. The standard assumption is stated by Luce's axiom [5], which assigns each item i on menu S a worth $u(i)$ with

$$p_i = \frac{u(i)}{\sum_{j \in S} u(j)}, \tag{1}$$

so the shares determine the worths up to scale. The prediction for a subset $T \subseteq S$ follows by changing the denominator alone, which leaves every ratio p_i/p_j with $i, j \in T$ as it was in S. For a two-item menu the predicted share preferring i is $p_i/(p_i + p_j)$. This is proportional renormalization. The invariance it imposes is Independence of Irrelevant Alternatives, and applied to a reduced set of response alternatives it is what Clarke [12] named the *constant-ratio rule.*

There is another approach that is equally natural, if not more so, and also demands no estimation or free parameters. Read the shares as the winning probabilities of a contest among

[2]The Gumbel construction is due to Holman and Marley [30, 28]; the uniqueness is Yellott's [34].

independent Gaussian performances, which is Thurstone's Case V [32] generalized from pairs to a multiway field. Item i carries a location a_i, draws $X_i \sim N(a_i, 1)$ on each occasion, and the highest draw is chosen, so

$$p_i = \int \varphi(x - a_i) \prod_{k \neq i} \Phi(x - a_k)\, dx. \tag{2}$$

Given the shares, this is inverted for the locations, which are identified up to a common translation and fixed here by $\sum_i a_i = 0$. The prediction for a subset T is the same contest run among T alone. For a pair it is one normal evaluation,

$$P(X_i > X_j) = \Phi\left(\frac{a_i - a_j}{\sqrt{2}}\right). \tag{3}$$

The two could be termed *linear normalization* and *Gaussian renormalization*. Both take the same shares and return shares over the survivors. The first holds the ratios of the shares fixed, the second holds the latent locations fixed.

The two horses disagree, and in a fixed direction. If $p_i > p_j$ then Equation (3) lies between $\frac{1}{2}$ and the renormalized $p_i/(p_i + p_j)$: winning a large field requires an unusually high draw and beating a single rival does not, so withdrawing the other alternatives counts the favourite's advantage for less. Section 3 shows that these two are the only members of the independent additive random-utility class costing nothing to apply, because fixing any other noise law requires a number the shares cannot supply, and Proposition 1 makes the direction exact.

2 Prior tests of the constant-ratio rule

Clarke tested the rule on spoken consonants in noise, predicting confusion matrices over reduced response sets from a larger one, and Clarke and Anderson [11] followed with a second test. Pollack and Decker [13] predicted four-way consonant matrices from an eight-way matrix and reported average deviations near four per cent. Hodge [15] ran nested ensembles in three modalities, Rich [16] carried the rule into short-term memory, and Getty, Swets, Swets and Green [14] held eight stimuli fixed while allowing only four responses.

Morgan [17] built a likelihood-ratio test and rejected the rule on the earlier data. Townsend and Landon [18] ran one master set of five letters with three nested response sets, blocked and counterbalanced, and published the full matrices per subject. Rouder [19] states the modern summary: conditioning on a reduced choice set is better than a guessing correction predicts and worse than the ratio rule predicts. Wills, Reimers, Stewart, Suret and McLaren [20] reject the rule in categorization and argue for a Thurstonian replacement.

Two findings recur across those studies. The rule is a fair first approximation, and it fails by concentrating the withdrawn mass onto whichever survivor most resembles the removed alternative rather than spreading it proportionally. Townsend and Landon attribute that to Debreu, and Rouder reaches the same diagnosis from the opposite direction.

What the literature does not contain is a comparison against a second assumption carrying no free parameters either. Lee [29] proposed one analytically, observing that detection theory and the constant-ratio rule diverge most for unidimensional stimuli and that the gap could serve in diagnosis, but did not carry it out on data. Nor was any of this scored out of sample; the tests are goodness-of-fit statements about the matrix in hand. Appendix B is a census of sixty-seven sources, recording which of them printed or deposited enough to score either assumption.

3 Two parameter-free maps

Renormalization is not a consequence of probability theory. It is the posterior for mutually exclusive events only when the information that one is impossible says nothing about which of the rest obtained, and Monty Hall is the familiar counterexample. Withdrawal is a third case, neither flat-likelihood conditioning nor informative conditioning, because no information about a fixed experiment has arrived. An alternative is gone and the problem is a different one. Conditioning requires a model of how the probabilities were generated, and renormalization supplies one.

Which model it supplies is known. Within the class of independent additive random-utility models, Gumbel errors generate Equation (1) [30, 28], and Yellott [34] showed the noise law is identified once menus of three or more are included, though not from pairwise comparisons alone. So the constant-ratio rule is one point of a family, and the empirical question is which point to stand on.

Equation (2) is a one-dimensional integral for any number of alternatives, since conditioning on $X_i = x$ makes the other events independent, and it is inverted numerically by the lattice method of [2], which recovers the whole location vector in one pass over a shared grid.

The class contains exactly two parameter-free points. Fix the noise law up to location and there is nothing left to choose. Gumbel gives renormalization, unit-variance Gaussian gives Case V, and every law between them costs a number. An analyst holding aggregate shares and no restricted observations therefore has two candidates rather than a family. Both are calibrated to the same $|S| - 1$ free shares and neither sees any observation from a smaller menu. Both are *representative-agent* maps, treating the population as one chooser with one set of worths or locations and transporting its shares from S to T. A fitted Plackett–Luce likelihood [31] uses the whole ranking and a fitted Bradley–Terry model [10] uses the pairwise outcomes directly; neither is estimated here.

The direction of the disagreement is a theorem, and it holds for noise laws other than the Gaussian. Let f and F be the density and distribution of the common noise, assumed independent and identically distributed across alternatives up to location, with positive density on a common support and $\log r$ differentiable, and let $r = f/F$ be its reverse hazard, with the highest draw winning as above.

Proposition 1 (Removing alternatives contracts the odds)**.** *Suppose* $\log r$ *is concave. Let* $T \subset S$ *and* $i, j \in T$ *with* $a_i > a_j$*. Then*

$$\frac{P(i \mid S)}{P(j \mid S)} \;\geq\; \frac{P(i \mid T)}{P(j \mid T)} \;\geq\; 1,$$

with the first inequality strict when $\log r$ *is strictly concave and* $S \setminus T$ *is non-empty.*

Proof. Write $A(x) = f(x - a_i)F(x - a_j)$ and $B(x) = f(x - a_j)F(x - a_i)$ for the two items in question, $C(x) = \prod_{k \in T \setminus \{i,j\}} F(x - a_k)$ for the other survivors, and $H(x) = \prod_{k \in S \setminus T} F(x - a_k)$ for the items removed. Then $P(i \mid T) \propto \int AC$ and $P(i \mid S) \propto \int ACH$, and likewise for j with B in place of A. Now

$$\frac{A(x)}{B(x)} = \frac{r(x - a_i)}{r(x - a_j)},$$

which is increasing in x when $\log r$ is concave, since $a_i > a_j$ places $x - a_i$ below $x - a_j$ and $(\log r)'$ is decreasing. Under the probability measure proportional to $B(x)C(x)\,dx$ the functions

A/B and H are both increasing, hence positively correlated, so

$$\frac{\int ACH}{\int BCH} \geq \frac{\int AC}{\int BC},$$

which is the first inequality, strict under strict concavity when $S \setminus T$ is non-empty. For the second, raising a_i stochastically increases X_i and so raises $P(i \mid T)$ while lowering $P(j \mid T)$, and the two are equal at $a_i = a_j$ by exchangeability; hence $a_i > a_j$ gives $P(i \mid T) \geq P(j \mid T)$. $\square$

The proposition says that removing alternatives moves the ratio toward one.

Two laws sit at the ends of that condition. The Gaussian satisfies it strictly, since $\frac{d^2}{dx^2} \log r(x) = -\operatorname{Var}(Z \mid Z < x) < 0$. The Gumbel has $r(x) = s^{-1}e^{-x/s}$, so $\log r$ is affine, the inequality becomes an equality, and renormalization is recovered exactly, which is the Holman–Marley construction [30, 28]. The axiom is a boundary of the class, and it is where machine learning sits: a softmax over scores z_i is Equation (1) with $u(i) = \exp(z_i)$, so masking a softmax and rescaling the survivors is proportional renormalization.

The proposition gives the direction of the effect and not its size. Size depends on the shares as well as on the noise law, and recalibrating several laws to one share vector and then to another can reverse which of them contracts more, so tail thickness alone does not order the laws by how much they contract.

4 What is measured here

We revisit the question the constant-ratio-rule literature asked. Both maps are calibrated on full-menu shares alone, then scored by log loss on restricted menus neither has seen. Three things make that possible now. Trial-level archives are open, so restricted menus can be scored on the observations that produced them rather than on printed marginals. Inverting Equation (2) from shares alone is a single pass over a shared grid, so the Gaussian map costs nothing to apply. And a generative null is available: contraction of a noisily estimated share vector lowers log loss whether or not the axiom fails, so every gain below is reported net of what a population obeying the axiom exactly would produce on the same menus, which Section 13 details.

5 Scoreboard

Thirty-four comparisons follow, each a population whose full-menu shares calibrate both maps and whose restricted-menu choices score them. A comparison is a win for the race when its gain exceeds the gain a renormalizing population would produce, with a Monte Carlo tail frequency of 0.05 or less, a loss when renormalization exceeds the race on the same test, and a draw otherwise.

The losses are not scattered. All four are the same task, absolute identification of tones, and Section 14 argues they are one loss with four rows. The draws sit where they should, at the smallest ranking collection, the smallest consumer experiment and the two gamble frames whose shares are most concentrated. And the two largest wins, colour and symbol naming, come from the one experiment that also supplies its own control.

[1] Wikipedia has a negative raw gain and a positive excess: the race loses to renormalization in absolute terms while still beating what the axiom predicts of it. See Table 4.

6 Choices from real menus

Twelve of the thirteen collections used here are complete rankings, from which subset choice is read off rather than observed. One is not, and it comes first.

Costa-Gomes, Cueva, Gerasimou and Tejiščák [1] ran two experiments in which subjects chose separately from every non-singleton subset of five consumer goods, twenty-six menus; the first also included the five singletons. The archive names the goods by two-letter codes and we describe them no more precisely. Restriction is observed rather than imputed, which makes this the cleanest test in our collection, and the two experiments are separate studies with disjoint subjects, so we report them separately rather than pooled.

Subjects are split into five folds. On the training subjects, shares come only from the full five-item menu; both maps are calibrated to those shares and nothing else; each then predicts every menu of size two, three and four; and we score the choices actually made by held-out subjects. Intervals resample subjects and repeat the whole procedure, calibration included. Subjects who may defer are reported alongside the forced-choice subgroup, for whom no deferral coding enters. Of the 373 subjects, 364 record at least one usable choice from a non-singleton menu and enter the analysis.

The race is ahead in both experiments and in both treatments, and by more where deferral cannot interfere. The two experiments do not agree on strength. Experiment 1 clears its null comfortably, with Monte Carlo tail frequencies of 0.010 and 0.005; experiment 2, at fifty-eight per cent of the sample, does not, at 0.199 and 0.129. The direction replicates and the magnitude is not established by these data alone.

7 Menus revealed after the stimulus

The consumer experiment observes restriction but announces it: the subject sees the menu before deciding. Two perceptual experiments remove that confound, and one of them removes it by design.

Yeon and Rahnev [21] show a display whose dominant colour, among four, has to be named. In one condition the full menu is available. In a second the observer sees the identical display and is told only *after* it has gone that the answer is one of two named alternatives. Stimulus, exposure and observer are the same; the menu is not, and nothing about the smaller menu could have been anticipated while the percept formed. Their Experiment 2 repeats this with six symbols. Calibration uses that observer's full-menu row for that dominant item alone, and the two-alternative cells on the same observer and item are then predicted and scored.

The six-symbol gain is the largest here outside news slates. The same pairs were also run *announced in advance*, so the observer can aim attention at them. That arm was not analysed by the original authors and is recovered here from the trial files. There the advantage nearly disappears. Accuracy rises from 0.780 to 0.850, renormalization becomes almost exactly right, and the race begins to over-contract. So the race's advantage is not a property of these pairs, these colours or these observers. It appears when alternatives are genuinely withdrawn from a percept formed without knowledge of the menu. It does not appear when the menu was known in advance.

In both genuine restriction arms *both* maps over-predict the favourite: renormalization by 0.069 and the race by 0.055 at four alternatives, and by 0.096 and 0.064 at six. Contraction has the right sign and removes a fifth to a third of the error. Neither parameter-free default is calibrated here, and a fitted noise scale would beat both. That over-prediction is Yeon and Rahnev's own finding, reached with a fitted model, so their result is independent evidence that the correction runs toward contraction while bounding what either default can claim.

Utochkin, Azarov and Grigorev [22] supply the same structure in memory. An observer sees a studied photograph among three foils, or the same target against one named foil, with the same images in both arms. The race wins by +0.0037 per cell over 360 nested cells, interval [+0.0029, +0.0047], the tightest in the paper. The contraction is the same size against a foil from the target's own category as against a foil from another, which matters in Section 14: category membership is not what breaks independence.

8 Four further observed restrictions

Restriction is observed in four further populations, which differ in what does the removing.

A fourth was tried and discarded. Grocery stockout data records hourly sales by product, and a shopper may buy two items from one category, so the alternatives are not mutually exclusive and the object is a basket rather than a choice. Recovering choices would need transaction-level baskets, which the release does not contain.

Aguiar, Boccardi, Kashaev and Kim [9] observe choices from all thirty-one non-empty subsets of five lotteries with a default always available, so menus run from two to six alternatives, in a cross-section giving at most two disjoint menus per person. It is adversarial on the authors' own analysis, since they report that random utility cannot explain the population behaviour while a logit attention model survives.

The experiment was run three times on disjoint subject pools, differing only in how much arithmetic is needed to value an alternative: one operation, three, or five. The excess over the null rises with that requirement, from −0.0012 to +0.0045 to +0.0081, and the full-menu shares flatten alongside it, the largest falling from 0.357 to 0.257 against a six-alternative maximum entropy of 1.792.

We read this as the thesis working rather than as instability. When the arithmetic is one step the task has a computable answer and is closer to an examination question than to a preference judgement; subjects converge, the shares concentrate, and there is little preference left for any restriction map to describe. When it takes five steps nobody can compute the answer, so people fall back on what they actually prefer and the choice becomes psychological. A model of preference under noise should win in the second case and not the first. Winning in both, including where the answer is objectively determined, would be the result to distrust.

US presidential ballots supply restriction by statute. The alternatives are the same named candidates in every state and which of them appears varies with petition thresholds and filing deadlines, so the states are nested menus over identical alternatives. States are not interchangeable populations, so a state may only predict another whose two-party log odds are within 0.5.

News slates supply the largest sample and the only menus that are recorded rather than reconstructed. The Microsoft News dataset logs, for each impression, the exact set of articles displayed and which one was clicked [33]. Slates recur across users and one slate is frequently a subset of another, so click shares on the larger slate calibrate both maps and the choices made on the nested smaller one score them. Of 153,727 impressions, 846 slates recur ten or more times and yield 6,135 nested pairs. Because those pairs are drawn from a few hundred slates and overlap heavily, the interval below resamples slates rather than pairs.

Wikipedia links supply editorial removal. An article offers outbound links, a reader picks one exclusively, the destinations are the same articles from month to month, and editors change the set. Adjacent monthly clickstream dumps give the earlier month's shares and the later month's choices. The dump censors pairs below ten clicks, so a vanished destination may have been removed or may have fallen below threshold, and it records no link position, so layout is uncontrolled.

The news result is the largest here by an order of magnitude, on the only menus in the paper that were recorded at the moment of choice. Its weakness is assignment: the slates are chosen by a recommender, so a slate's composition is correlated with what the system expects readers to want, and a smaller slate may reach a different readership than the larger one. Both maps inherit that identically and the null runs on the same slates, which is what makes the comparison fair without making the assignment random.

The lottery gradient runs with arithmetic cost, and the mechanism of Appendix A predicts its direction, since computation error is per-alternative and roughly Gaussian, so loading more of it onto each alternative raises σ against β. And the Wikipedia rows say something the ranking collections cannot: tightening the filter that identifies a genuine removal from a five per cent share to thirty-five per cent raises the excess from +0.0001 to +0.0005 across 3,544, 817 and 121 pages, so the near-zero headline is attenuation by measurement rather than absence of an effect. Both Wikipedia and the two General Social Survey collections put the truth between the two defaults: choice contracts when an alternative goes, but by less than Case V says.

9 Choices induced from rankings

Twelve archival collections have human respondents and rankings, or ratings fine enough to yield them. Two rank-order card tasks from the General Social Survey ask what a child should learn and what matters in a job. Three independent samples of Jester users rate the same ten jokes. Sushi rankings come from a Japanese web survey, film rankings are induced from Netflix ratings, and dots and puzzles are perceptual discrimination tasks. The rest are rankings of seven sports by participation preference, ten occupations by perceived prestige, and four political goals from a five-nation survey. The thirteen collections comprise 147,719 responses, of which the menu experiment contributes 373 subjects; the twelve ranking collections contribute 147,346. These are not thirteen independent experiments, since three are Jester files, two are General Social Survey items, and dots and puzzles are companion tasks.

The protocol matches Section 6 with respondents in place of subjects, and the outcome for a subset is the respondent's highest-ranked member of it.

10 Gain by size of the surviving menu

Splitting the held-out predictions of Section 9 by the size of the surviving menu shows the gain concentrated where most of the menu has been withdrawn.

11 The older datasets rescored

Four subjects of Townsend and Landon [18] identified briefly displayed block capitals, in blocks with the master response set $\{A, E, F, H, X\}$ and blocks with $\{A, E, F, H\}$, $\{A, E, X\}$ and $\{F, H, X\}$, counterbalanced, 240 trials per row. Responses were spoken letter names, so the labels are identical across menus by construction. They published the full confusion matrices per subject, which is why the dataset is still usable forty-four years later.

It is the cleanest calibration in the paper for a reason none of the others share. Calibration row and target row come from the *same subject* in different blocks, so there is no population-heterogeneity confound of the kind every ranking collection carries, and the aggregation argument of Appendix A is not available as an explanation of any gain found here.

Over 38 restricted rows and 9,120 held-out trials, renormalization scores 0.9454 and the race 0.9412, a gain of $+0.0042$ with a row bootstrap of $[+0.0022, +0.0064]$, against a fitted-Luce null median of -0.0011, so the excess is $+0.0053$ at $p = 0.005$. The subsets give $+0.0006$ $[-0.0002, +0.0015]$ for $\{A, E, F, H\}$, $+0.0040$ $[-0.0003, +0.0079]$ for $\{A, E, X\}$ and $+0.0093$ $[+0.0051, +0.0141]$ for $\{F, H, X\}$. That ordering is mostly the removal count: the first subset withdraws one letter and the other two withdraw two, and Table 6 shows the gain growing with how much is removed and vanishing when nothing is. The unconfounded comparison is between the two subsets that each remove a similar pair and retain the other, $+0.0040$ against $+0.0093$. Those two designs are structurally symmetric, each removing a similar pair and retaining the other, so the difference between them is not attributable to similarity.

What the row-level output does show is the limit of a parameter-free correction. For subject one, stimulus A, subset $\{A, E, X\}$, the observed, renormalized and raced shares are 0.621/0.672/0.635 on A, 0.217/0.129/0.149 on E and 0.163/0.199/0.217 on X. The race corrects the favourite downward and the near-substitute upward, both in the right direction and the second far too little, and moves the dissimilar letter the wrong way. It wins because the dominant terms improve. Townsend and Landon diagnosed their residual as mass concentrating onto the surviving near-substitute instead of spreading, and Case V has no covariance to represent that, so the residual survives. A parameter-free Gaussian recovers part of the discrepancy and does not remove it. Lee [29] proposed exactly this comparison analytically in 1968, observing that detection theory and the constant-ratio rule diverge most for unidimensional stimuli and that the gap could serve in diagnosis; what is added here is the out-of-sample follow-through.

Wills et al. (2000)

Wills et al. deposited their Experiment 2, in which twelve participants chose among three categories and twelve others had one category disallowed, over the same stimuli. Over 39 cells and 1,560 held-out two-choice trials, renormalization scores 0.6839 and the race 0.6540, a gain of $+0.0299$ against a null median of -0.0015, so the excess is $+0.0314$ at $p = 0.002$. Graded and dummy stimuli agree, $+0.0287$ against $+0.0326$.

The restriction is between groups, with four participants behind each disallowed category, and the gain sits in two of the three: -0.0001, $+0.0413$ and $+0.0485$. The participant bootstrap is wide and right-skewed, $[+0.0175, +0.0717]$.

The withdrawn response occurs zero times in the two-choice condition and 510 times in the three-choice condition, so it is a real competitor on the master menu, and cells are balanced at forty trials.

12 Where the difference lives

Averaged over outcomes the difference between the two defaults is small, and averaging hides what it is made of. Score each held-out prediction by the rank the chosen item held in the full-menu shares, pooling the four ten-item collections:

The race pays 0.062 nats whenever the favourite wins, which happens 17 per cent of the time, and collects 0.171 when the tenth-ranked item wins, which happens 4 per cent of the time. That is what a contraction does: it moves probability off the favourite and onto the tail, buying insurance against upsets and paying a premium on every favourite win. Three earlier observations follow. The aggregate gain is small because large rare wins net against small frequent losses. The ballot result is negligible because third-party candidates are the tail and there is almost no tail mass to collect. And the effect is largest on news slates, where readers spread widely over the displayed set.

A second decomposition names an error rather than a difference. Ask how often a given item finishes second, which is the conditional question after the winner is removed:

This is the pattern reported for the Harville formula in racing place markets, favourites over-priced and outsiders under-priced, appearing in survey rankings of sushi, jokes, child-rearing values and political goals. It is the most useful result here for a practitioner, because it is a large, replicable, signed error in the default rather than a small average edge. The race removes 5 to 12 per cent of it, so both rules share a bias much larger than the distance between them. What produces the remainder is not identified; the obvious candidate is that a single respondent's ranking is internally consistent, so adjacent places are positively correlated in a way no independent contest reproduces.

13 Shrinkage

A positive gain has two possible sources and log loss cannot tell them apart. The race may be closer to how the population chooses. Or the race may be wrong about that and win anyway, because contracting a noisily estimated share vector is shrinkage, and shrinkage lowers log loss when the input is over-confident. The second mechanism needs no departure from the axiom, so properness of the score licenses nothing here: a proper score says the true probability vector is optimal, not that a plug-in estimate of it beats a biased transformation of the same estimate.

The remedy is to build a population where the axiom holds by construction and run the identical procedure on it. Taking the observed full-menu shares as worths, choices or complete rankings are generated from that exact process, and everything is repeated. Rankings are drawn by sorting $\log u_i + G_i$ for independent standard Gumbel G_i, which is the Plackett–Luce ranking law; for $K \geq 4$ this is a particular joint law consistent with those subset marginals rather than the only one, so the null is specifically a Plackett–Luce ranking null.

The shrinkage mechanism is real and it accounts for the entire gain in the smallest dataset. Sports participation, with 130 respondents, sits exactly on its null and is excluded from every claim here. Everywhere else the mechanism runs against the race, and the excess over the null exceeds the raw gain in all twelve remaining rows. It also rescues puzzles, whose raw interval in Table 5 covers zero while its excess over the null does not. The menu experiment clears the null on both readings, which is the result the paper rests on, since it is the only one where restriction is observed.

14 Where the race loses

The four losses in Table 1 are one task, and three further populations fail both maps in ways that published aggregates settle without this protocol. Taken together they describe a boundary that can be stated as a rule rather than a list.

A perceptual continuum. Stewart, Brown and Chater [23] had listeners name one of ten pure tones, and ran the same tones with the middle eight and the middle six labels, so the label sets nest over physically identical stimuli. Calibrating on the ten-label row and predicting the restricted rows, renormalization wins all four conditions, by 0.0133 and 0.0051 nats at narrow spacing and 0.0173 and 0.0057 at wide. The mechanism is visible in the matrices, which are strongly banded: a tone is confused almost entirely with its immediate neighbours, so the alternatives are near-substitutes by construction and removing the outer labels returns their mass to neighbours rather than proportionally. The loss is larger at six labels, where more of the continuum has been cut away.

A survivor's near-substitute. Scottish juries return one of three verdicts, and Ormston and colleagues [26] ran 64 mock juries on films that were identical except for the final section, in which the judge names the verdicts available. Half had three verdicts and half had two. Renormalizing the three-verdict shares onto the survivors predicts Guilty at 54.9 per cent against an observed 38, and after deliberation 61.1 against 31. Both defaults fail, because the ordering *reverses*: Guilty leads Not guilty with three verdicts and trails it with two. Contraction moves odds toward one and never crosses over, so the race fails here too, and only slightly less. Not proven is a near substitute for Not guilty, so deleting it returns its mass to that neighbour rather than proportionally, which is Debreu's objection in a courtroom, on a fixed stimulus. Townsend and Landon's diagnosis [18] is the same one: their rule fails on the subset containing similar letters and holds on the subset that does not.

The same boundary inside one experiment. Getty, Swets, Swets and Green [14] had three observers identify eight complex sounds with all eight responses available, and then with only four, a different four in each of three conditions, the labels being the stimulus numbers. The stimulus set never changes; only the menu does. Over 72 cells the race wins by $+0.0272$ nats, and by $+0.0111$ on the rows whose stimulus survives in the menu, cell bootstrap $[+0.0021, +0.0225]$. But it loses one condition, and which one is not arbitrary.

The statistic turns the near-substitute condition from a description applied after a failure into a quantity computable in advance. Townsend and Landon [18] were using the same diagnostic in 1982, when they reported their rule holding on one subset and failing on the subset containing similar letters. With three observers the bootstrap resamples cells rather than observers and understates uncertainty, so only the signal-row interval excludes zero. The authors also report that observers retuned the weights they placed on perceptual dimensions to maximise discriminability of whichever subset they had to identify, with feedback given only on that subset, so both boundary conditions are present in this experiment at once.

Removal that changes the task. McMurray and colleagues [24] ran a two-label phoneme identification and a four-label version nesting it, on identical synthetic stimuli. The two extra labels absorbed under one per cent of responses, so renormalization predicts almost no change and contraction predicts movement toward even; the observed identification slope instead grew steeper, which is neither. Zoanetti and colleagues [25] report the same direction in assessment: deleting a distractor raised the share choosing the correct answer by more than proportionally. In both cases the survivors became easier to tell apart, so they are not the same alternatives before and after, and no map from old shares to new ones can be right.

Shares too concentrated to discriminate. Where one alternative holds most of the mass the two defaults agree to within the noise, and the comparison carries no information either way. The low-cost gamble frame, the ballot rows and the Wikipedia rows are all of this kind, which is why they are draws or near-zero wins rather than evidence.

The rule that summarises this is narrower than the class of independent random utility models and wider than any single dataset. *The Gaussian contest is the better default where the alternatives are distinct unordered items*, such as goods, news articles, memory foils, sushi, jokes, ballots, colours and symbols. It is not where they lie on a perceptual continuum, where a removed option is a near-substitute for a survivor, or where removal changes the discriminability of what remains.

Two consequences follow. First, an apparent conflict resolves. Meyer-Grant and colleagues [27] show that Gumbel-min uniquely predicts accuracy invariance as a recognition set grows uniformly, find that invariance, and report the Gumbel model outperforming the Gaussian throughout. That is Yellott's own condition, independently confirmed, in the same domain where Section 7 has the race winning on nested foils. Both hold, and their footnote supplies the reconciliation: invariance breaks down when systematic similarity is introduced, because latent strengths cease to be independent. Our foil split found contraction equal across same- and cross-category foils while the tones show similarity mattering greatly, which says it is the continuum that breaks independence and not category membership. Two photographs of different dogs are far less confusable than two tones six per cent apart in frequency.

Second, the obvious repair is unavailable as a default. Correlated multinomial probit would handle the near-substitute cases exactly, by giving Not proven and Not guilty a positive covariance. But the analyst here holds $K-1$ free shares and the model has $K(K-1)/2-1$ free covariance parameters, so for $K \geq 3$ the shares cannot identify them. A correlated model is available to anyone who has restricted-menu data to fit it on, and that is a different exercise. The boundary therefore marks the conditions under which no parameter-free default can work, rather than a gap awaiting a better one.

15 Scope

The practical recommendation is narrow, and Section 14 makes it conditional. When the task is to infer restricted-menu population shares from full-menu population shares, *and the alternatives are distinct unordered items*, the Gaussian race should be the default in place of renormalization. Four questions decide whether that condition holds, and all four can be answered before any restricted-menu data exists. Do the alternatives lie along a single physical or perceptual dimension, so that neighbours are more confusable than distant pairs? Is any alternative a near-substitute for another, in the sense that a chooser who loses one would take the other rather than spread? Does removing an alternative change how easily the survivors can be told apart, as deleting a distractor or an implausible response option does? And is one share already so large that the two maps agree to within the noise? A yes to the first three says renormalization or a fitted correlated model, not the race. A yes to the fourth says the choice does not matter. A banded confusion matrix answers the first two on inspection, which is the diagnostic Townsend and Landon [18] already used.

The inversion from shares to locations is the only extra machinery, and the lattice algorithm of [2] performs it in one pass over a common grid, returning the whole vector of winning probabilities at once. The restricted predictions are then one-dimensional quadrature on the same grid. At the field sizes here, ten alternatives at most, both steps are immediate; a field of $n = 10{,}000$ takes a fraction of a second, and the method scales to fields of millions [2]. So the recommendation needs no additional data and no appreciable additional computation. Within its condition it was better on every population examined here; outside it, on the tone matrices, it was worse on all four.

All twelve ranking collections carry an assumption the menu experiments do not. A respondent's ranking imposes one ordering on every subset by construction, so no ranking dataset can show a person choosing i over j from one menu and j over i from another, and none reveals whether that person would choose as their ranking implies when handed two options. Reading restriction off a ranking assumes they would. Section 6 does not.

Every quantity here is a population quantity. A population of people who each choose by Equation (1), with worths differing between people, generally does not satisfy the axiom in aggregate, which is why mixed logit is useful and why its probabilities are integrals over logit probabilities rather than logit probabilities [7]. Aggregate failure of the axiom is therefore not evidence that individuals violate it. Heterogeneous Luce choosers can reproduce aggregate contraction, and Appendix A exhibits a class in which they do. What is rejected is a representative-agent map.

The experiment compares two restriction maps, not two fitted likelihoods. A fitted Plackett–Luce model [31] uses the whole ranking and a fitted Bradley–Terry model [10] uses the pairwise outcomes directly. Neither was estimated here. Recommendations about ranking or paired-comparison likelihoods are therefore outside what these tables measure.

The score is a sum of marginal scores for top-item-within-subset outcomes. It is proper for those marginals and not for the ranking distribution. With $K = 4$ the scored marginals carry at most 17 degrees of freedom against 23 for a distribution over rankings, so distinct ranking laws can agree on everything scored here.

The independent-noise family nests Luce at the Gumbel law, so it excludes no Luce population. Choosing the Gaussian point of it is a separate decision, and Table 9 shows that decision losing when the axiom holds and the shares are well estimated.

Neither map dominates the other. When the shares are precise the better map is whichever matches the population, and when they are very noisy the dominant consideration is estimation risk rather than either structural rule. The recommendation above is for the case where an analyst holds aggregate shares and has no evidence about the axiom either way.

One limit is larger than the effect measured here. Both defaults can be checked against the truth, rather than only against each other, in the after-the-stimulus arms of Section 7 and in the favourite's second-place probability of Table 8. In both, they are wrong in the same direction and by more than they differ from one another. The race removes a fifth to a third of renormalization's error in the first case and five to twelve per cent of it in the second. Choosing the better of two parameter-free maps is therefore a smaller improvement than fitting one number would be, wherever a restricted-menu observation exists to fit it on. The case made here is for the situation where none does.

Pricing ordered outcomes by removing the winner and re-running is not the race's own prediction. Against the calibrated race's ordering law it misprices individual pairs and triples by up to a factor of 3.4. We leave to future work the comparison of the exact Gaussian ordering law against Plackett–Luce ranking probabilities on complete-ranking data.

Data and code

Sushi rankings come from Kamishima [4]; Jester ratings from the Eigentaste project [3]; the General Social Survey cumulative file from NORC; the perceptual, leisure, prestige, film and political-goal rankings via PrefLib [6] and the CRAN packages PLMIX and ConsRank; the menu-choice data from the replication archive of [1]; the colour and symbol naming trials from the OSF archive of [21], with the advance-warning condition recounted from the raw session files because the authors' aggregate matrices omit it; the recognition trials from the OSF archive of [22]; the tone confusion matrices digitized from the vector figures of the author manuscript of [23]; the jury shares from the published tables of [26]. All are public. Analysis code and every input needed to reproduce the tables are under `research/restriction` in the repository `github.com/microprediction/winning`, the inputs as column extracts rather than raw archives; an archival snapshot with a DOI accompanies circulation.

Two storage conventions must be told apart when reproducing this. BayesMallows, ConsRank and PerMallows store ranks, so element j holds the rank of alternative j, while PLMIX stores orderings, so the first element names the favourite. Reading one as the other transposes the data silently and raises no error. We established which is which by cross-validation: the political goals data ships in two of these packages and they agree only when each is read under its own convention.

A Why a population should look Gaussian

Nothing so far explains why the Gaussian member of the class should be the one that fits. A mechanism is available, and it is elementary.

Let every individual obey the axiom, and let their tastes differ. Individual θ assigns systematic utility $V_i(\theta)$ to alternative i and chooses by Luce at scale $\beta > 0$,

$$P(i \mid S, \theta) = \frac{\exp\{V_i(\theta)/\beta\}}{\sum_{j \in S} \exp\{V_j(\theta)/\beta\}}, \tag{4}$$

which by Yellott's theorem is the same as choosing the largest $V_i(\theta) + \beta G_i(\theta)$ for independent standard Gumbel $G_i(\theta)$.

Proposition 2 (Gaussian taste heterogeneity)**.** *Suppose $V_i(\theta) = \mu_i + \sigma Z_i(\theta)$ with $Z(\theta) \sim N(0, I_K)$ independently across individuals. Then the aggregate choice probabilities are exactly the winning probabilities of a contest with utilities*

$$\mu_i + \sigma Z_i + \beta G_i, \tag{5}$$

independent across alternatives. If $\mu_i = \sigma a_i$ and $\sigma/\beta \to \infty$, they converge to those of Case V with locations a.

Proof. Aggregating over individuals integrates over the taste draw Z, leaving the per-alternative utility (5), independent across i because both components are. Dividing every utility by σ preserves the arg max and gives $a_i + Z_i + (\beta/\sigma)G_i$, whose last term vanishes in probability. Ties have probability zero under the Gaussian limit, so the winning probabilities converge. □

This is a mixed logit: Gaussian heterogeneity between individuals, Gumbel noise within them. Such models are standard, and McFadden and Train [7] establish how general they are.

Nothing here claims otherwise. What the proposition supplies is the connection between the high-heterogeneity limit of that familiar object and the particular restriction map the paper tests.

The Gaussian assumption on taste is where a limit theorem belongs.

Corollary 1 (Many small taste components)**.** *Suppose $V_i(\theta) = \sqrt{m}\, a_i + \sum_{\ell=1}^{m} \xi_{\ell i}(\theta)$, where the vectors $\xi_\ell(\theta)$ are independent and identically distributed across ℓ with mean zero, covariance I_K and finite second moments, and the individual chooses by (4) at fixed β. Then the aggregate choice probabilities converge as $m \to \infty$ to those of Case V with locations a.*

Proof. Dividing the utilities by $\sqrt{m}$ gives $a_i + m^{-1/2} \sum_\ell \xi_{\ell i} + (\beta/\sqrt{m}) G_i$. The multivariate central limit theorem sends the middle term to $N(0, I_K)$ and the last to zero, and the arg max probabilities are continuous at a limit with no ties. ☐

One direction for future work would turn this limit into a correction. At finite σ/β the composite noise is $Z + \varepsilon G$ with $\varepsilon = \beta/\sigma$, so the natural device is a cumulant expansion about the Gaussian rather than anything singular. The first two Gumbel cumulants do little: the mean is a common location shift and cancels in the arg max, and the variance inflation $1 + \pi^2 \varepsilon^2/6$ is a common scale change, which would be absorbed entirely if the calibration matched the whole law. It does not. It matches the $K-1$ full-menu shares, so a common scale error is invisible in the fit and propagates into the restricted-menu prediction, which is the quantity at issue. Simulating the deviation of observed contraction from the recalibrated Case V value over $\varepsilon \in [0.125, 1]$ gives local exponents of 1.6, 2.0 and 2.3, so the range we can resolve looks quadratic rather than cubic; the smallest point sits near Monte Carlo resolution and deterministic quadrature would be needed to settle the rate. Either way the target is already tabulated, since the ratio column above is what such an expansion must reproduce.

The covariance assumption is doing real work and should not be waved through. If the taste components carry covariance Σ instead of I_K, the same argument delivers a Gaussian race with covariance Σ, which is correlated multinomial probit rather than Case V. Case V requires the covariance relevant to utility differences to be spherical. So the limit theorem motivates Gaussian random utility in general, and independence across alternatives is a further restriction, adopted here because it is the parameter-free member of the class.

No individual need be non-Lucean for the Gaussian transport to be the right one in aggregate. A population of Luce choosers whose tastes vary is a contest with composite noise, and as taste variation comes to outweigh within-person choice noise the composite approaches Gaussian and the population contraction approaches the Case V prediction. That mechanism also says where the fit should be worst: in populations whose members are nearly alike.

Independent evidence for the same phenomenon exists in memory research. Robinson and colleagues [8] compare softmax against signal detection across m-alternative tasks and report that the softmax parameter must vary with m while the signal-detection parameter is stable. That is the contraction studied here, seen through the parameter that has to move to absorb it.

B Census of restriction experiments

The experiments exist. They were run in four literatures that do not cite one another, speech confusion, visual absolute identification, categorization and odor identification, and most of them published percent correct rather than cell-level shares, which scores neither map. That describes the majority of what we found.

We assembled a census of sixty-seven sources, one note each, recording design, what numbers were printed or deposited, a fetched access route, and a usability verdict, negatives included. It is in the repository under `research/restriction/notes/crr`. Table 12 lists the sources that restrict a response set over shared alternatives, are not scored above, and could therefore still change the answer.

None of the eleven entries is a preference task, so the corpus scored above is not representative of this literature by domain. The one entry needing only transcription is the oldest, published in 1960.

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Population	Restriction	Tail	Verdict
Occupational prestige	ranking-induced	$\leq .005$	win
Sushi	ranking-induced	$\leq .005$	win
Political goals	ranking-induced	$\leq .005$	win
GSS socialization	ranking-induced	$\leq .005$	win
GSS job values	ranking-induced	$\leq .005$	win
Jester files 1, 2, 3	ranking-induced	$\leq .005$	win $\times 3$
Dots	ranking-induced	$\leq .005$	win
Puzzles	ranking-induced	$\leq .005$	win
Netflix	ranking-induced	$\leq .005$	win
Sports participation	ranking-induced	.51	draw
Colour naming, four to two	observed menus	$\leq .005$	win
Symbol naming, six to two	observed menus	$\leq .005$	win
Recognition foils, four to two	observed menus	$\leq .005$	win
News slates	observed impressions	.016	win
Consumer goods, experiment 1	observed menus	.005	win
Gambles, high arithmetic cost	observed menus	.035	win
US state ballots 2024	observed ballots	.005	win
Letters, five to three, $\{F, H, X\}$	observed menus	.005	win
Categories, three to two	observed menus	.002	win
Letters, five to three, $\{A, E, X\}$	observed menus	.010	win
Sounds, eight to four, spread sets	observed menus	$\leq .005$	win $\times 2$
Wikipedia links	observed removals	.005	win[1]
Consumer goods, experiment 2	observed menus	.129	draw
Gambles, medium arithmetic cost	observed menus	.124	draw
Gambles, low arithmetic cost	observed menus	.667	draw
Letters, five to four, $\{A, E, F, H\}$	observed menus	.129	draw
Sounds, eight to four, similar set	observed menus	1.000	loss
Tones, narrow spacing, ten to six	observed menus	n/a	loss
Tones, narrow spacing, ten to eight	observed menus	n/a	loss
Tones, wide spacing, ten to six	observed menus	n/a	loss
Tones, wide spacing, ten to eight	observed menus	n/a	loss

Table 1: Twenty-four wins, five draws, five losses. The first block reads subset choice off a complete ranking; the second and third observe it. Tail frequencies are floored at 1/201 by two hundred replicates. The tone matrices are participant-averaged and between-subjects, so there is nothing to resample and the losses are point comparisons. Forced-choice subgroups are quoted for the consumer experiments, since no deferral coding enters those. Three further populations appear in Section 14 rather than here, because published aggregates fix the direction of the failure without supporting this protocol.

	n	Choices	Renorm.	Race	Gain	Excess
Experiment 1, all	230	5,134	0.9438	0.9298	+0.0140	+0.0158
Experiment 1, forced	76	1,900	0.9639	0.9197	+0.0442	+0.0466
Experiment 2, all	134	3,105	0.9162	0.9103	+0.0059	+0.0056
Experiment 2, forced	61	1,525	0.8939	0.8799	+0.0140	+0.0131
Pooled	364	8,239	0.9263	0.9164	+0.0100	+0.0123

Table 2: Held-out log loss per choice on menus of size two to four, both maps calibrated only to training-fold full-menu shares under a common add-α convention, $\alpha = 1/2$. Excess is the gain net of the median gain under a population that renormalizes exactly, as in Section 13. Subject-bootstrap intervals for the gains are $[+0.0072, +0.0335]$, $[+0.0171, +0.0867]$, $[+0.0009, +0.0306]$, $[+0.0034, +0.0560]$ and $[+0.0048, +0.0246]$ respectively.

Arm	Cells	Renorm.	Race	Gain	Excess
Four colours, menu revealed after	384	0.4942	0.4795	+0.0147	+0.0157
Six symbols, menu revealed after	300	0.6276	0.5999	+0.0278	+0.0314
Same pairs, announced in advance	384	0.3967	0.3928	+0.0038	+0.0048

Table 3: Mean nats per cell, a cell being one observer, one dominant item and one surviving alternative. Subject-bootstrap intervals for the gains are $[+0.0118, +0.0179]$, $[+0.0234, +0.0324]$ and $[+0.0006, +0.0072]$. The null here resamples the calibration row as well as the pair counts, so it charges the race for calibration noise as well as for sampling noise in what it predicts; all three tails are at the 0.005 floor.

Population	Units	Gain	Null median	Excess	MC tail
Lotteries, high cost	4,099	+0.0101	+0.0020	+0.0081	.035
Lotteries, medium cost	4,099	+0.0068	+0.0023	+0.0045	.124
Lotteries, low cost	4,099	+0.0002	+0.0014	−0.0012	.667
News slates	6,000 pairs	+0.0496	−0.0088	+0.0584	.016
US state ballots 2024	263 pairs	+0.0002	−0.0000	+0.0002	.005
Wikipedia links	3,544 pages	−0.0001	−0.0002	+0.0001	.005

Table 4: Held-out loss differences on populations where restriction is observed rather than read off a ranking. Excess is the gain net of what a renormalizing population produces on the same menus. The lottery frames are disjoint subject pools; ballot units are matched state pairs; Wikipedia units are pages whose destination set shrank; Two rows have a negative raw gain and a positive excess, which means the race loses to renormalization in absolute terms while still beating what the axiom predicts of it.

Dataset	n	K	Subsets	Renorm.	Race	Gain
Occupational prestige	143	10	1,013	1.0173	0.9783	+0.0390 [+0.0227, +0.0555]
Sushi	5,000	10	1,013	1.3724	1.3613	+0.0111 [+0.0094, +0.0128]
Political goals	2,262	4	11	0.8835	0.8766	+0.0069 [+0.0059, +0.0078]
GSS socialization	5,000	5	26	0.7984	0.7929	+0.0055 [+0.0041, +0.0070]
Jester file 3	5,000	10	1,013	1.5189	1.5165	+0.0024 [+0.0015, +0.0032]
Jester file 1	5,000	10	1,013	1.5287	1.5267	+0.0020 [+0.0012, +0.0027]
Jester file 2	5,000	10	1,013	1.5247	1.5230	+0.0018 [+0.0010, +0.0025]
GSS job values	5,000	5	26	0.8593	0.8576	+0.0017 [+0.0005, +0.0030]
Dots	3,183	4	11	0.8285	0.8270	+0.0015 [+0.0005, +0.0025]
Netflix	4,379	3	4	0.7932	0.7922	+0.0009 [+0.0007, +0.0012]
Puzzles	3,180	4	11	0.8180	0.8173	+0.0007 [−0.0003, +0.0017]
Sports participation	130	7	120	1.2579	1.2528	+0.0050 [+0.0017, +0.0087]

Table 5: Held-out log loss per prediction over every subset of size two or more. Datasets are capped at five thousand respondents for runtime. The race is lower in twelve of twelve and the interval excludes zero in eleven. Occupational prestige is scorable here only because the shared smoothing removes a zero training cell. Sports participation sits below the rule: its gain is real but Section 13 shows it carries no evidence about the restriction map. Intervals resample respondent losses with the fitted training models held fixed, so unlike Table 2 they omit calibration uncertainty and are too narrow, most seriously in the two smallest rows.

Dataset	$\lvert T\rvert = 2$	3	4	5	6	7	8	9	10
Sushi	+.0412	+.0263	+.0150	+.0081	+.0040	+.0017	+.0005	+.0000	−.0000
Jester file 3	+.0064	+.0050	+.0033	+.0020	+.0012	+.0006	+.0003	+.0001	−.0000
Jester file 1	+.0053	+.0041	+.0026	+.0016	+.0010	+.0005	+.0003	+.0001	+.0000
GSS socialization	+.0130	+.0011	+.0004	−.0000					
Political goals	+.0079	+.0071	+.0000						
Dots	+.0025	+.0003	−.0000						
Netflix	+.0013	+.0000							

Table 6: Held-out gain by the size of the surviving menu, for a representative seven of the twelve ranking collections. The pairwise gain is two to four times the all-subsets aggregate, and the last entry in each row is exactly zero because nothing has been removed and the two maps must coincide. The aggregate quoted elsewhere is therefore lower than the pairwise figure. Occupational prestige is the one non-monotone row and is omitted here.

Rank of chosen item	1	2	3	4	5	6	7	8	9	10
Share of outcomes	.17	.13	.13	.10	.10	.10	.08	.07	.07	.04
Mean gain	−.062	−.045	−.034	−.004	+.009	+.019	+.050	+.058	+.101	+.171

Table 7: Held-out gain by the full-menu rank of the item actually chosen, pooled over Sushi and the three Jester files. Monotone in rank, crossing zero between fourth and fifth. The contributions sum to +0.0043.

Collection	Observed	Renormalization	Race
Political goals	0.123	0.318	0.309
GSS job values	0.186	0.303	0.294
GSS socialization	0.191	0.302	0.294
Sushi	0.206	0.251	0.245
Jester file 1	0.138	0.160	0.158

Table 8: Probability that the full-field favourite finishes second. Both columns are sequential rules: remove the winner, then apply the map to the survivors. That is Harville's construction in the first column and its re-run-the-contest analogue in the second, and neither is the exact ordering law of its model, so this compares two practical rules rather than two models. Renormalization overstates the probability in every collection, by 0.022 to 0.195; the race overstates it too, by a little less. The mirror holds for the last-ranked item, whose chance of finishing second both rules understate everywhere.

Dataset	Observed	Null median	Excess	MC tail
Menu experiment, forced choice	$+0.0265$	-0.0041	$+0.0306$	0.010
Menu experiment, all subjects	$+0.0100$	-0.0023	$+0.0123$	0.015
Occupational prestige	$+0.0390$	-0.0227	$+0.0617$	≤ 0.005
Sushi	$+0.0111$	-0.0062	$+0.0174$	≤ 0.005
GSS socialization	$+0.0055$	-0.0066	$+0.0121$	≤ 0.005
Political goals	$+0.0069$	-0.0012	$+0.0081$	≤ 0.005
GSS job values	$+0.0017$	-0.0047	$+0.0065$	≤ 0.005
Jester file 3	$+0.0024$	-0.0015	$+0.0039$	≤ 0.005
Jester file 1	$+0.0020$	-0.0014	$+0.0034$	≤ 0.005
Jester file 2	$+0.0018$	-0.0014	$+0.0032$	≤ 0.005
Dots	$+0.0015$	-0.0014	$+0.0029$	≤ 0.005
Puzzles	$+0.0007$	-0.0011	$+0.0019$	≤ 0.005
Netflix	$+0.0009$	-0.0001	$+0.0010$	≤ 0.005
Sports participation	$+0.0050$	$+0.0051$	-0.0000	0.51

Table 9: The same gain when the axiom holds by construction. Two hundred replicates per row. The diagnostic is the excess of the observed gain over the null median; the tail probabilities are indicative rather than exact tests, and for the menu rows they are optimistic, because the null redraws each subject's thirty-one choices independently given the aggregate worths and so omits the within-subject dependence that the bootstrap intervals of Table 2 do preserve. Wherever shares rest on more than about two thousand observations the null is negative, because contraction is then a bias rather than useful insurance, so comparing an observed gain with zero understates it.

Condition	Survivors	Within-set confusion	Gain
1	$\{1, 2, 5, 6\}$	0.103	+0.0453
3	$\{1, 3, 5, 7\}$	0.335	+0.0490
2	$\{3, 4, 5, 6\}$	0.790	−0.0127

Table 10: Within-set confusion is the fraction of a surviving stimulus's errors on the *full* eight-way menu that land on another survivor of that condition. It uses the master matrix only, so it is available before any restricted-menu observation exists. The one condition the race loses is the one whose survivors are each other's confusions.

σ	Top share	Observed λ	Case V λ	Ratio
0.5	0.255	0.052	0.225	0.23
1	0.300	0.128	0.227	0.57
2	0.343	0.197	0.229	0.86
3	0.359	0.216	0.229	0.94
4	0.365	0.228	0.230	0.99
8	0.373	0.230	0.231	1.00
16	0.375	0.233	0.231	1.01

Table 11: A population of Luce choosers whose log-worths carry Gaussian taste variation of scale σ against a Gumbel choice-noise scale of 1, with locations scaled as σa so that the shares do not degenerate. Observed λ is the population's contraction; Case V λ is what the independent unit-variance Gaussian race implies for that population's own shares. The top share is reported to show the limit is not the near-indifference corner.

Source	Domain	Restriction	Status
Clarke [12]	speech in noise	$8 \to 4$, $6 \to 3$	library access
Pollack and Decker [13]	consonants	$8 \to 4$, three sets	printed; digitize
Hodge and Pollack	auditory, unidimensional	nested	library access
Lacouture et al.	line lengths	nested	library access
Kent and Lamberts	dot separation	$N = 2, 4, 6, 8, 10$	library access
Rouder lab, unpublished	line lengths	twelve, nested	raw trials, open
Negoias et al.	odor descriptors	$16 \to 8 \to 4$	never published
Jäger et al.	flavour terms	nested, $n = 735$	library access
Hörberg et al.	odor naming	free, then cued list	data request

Table 12: Restriction experiments not scored above. "Printed" means the matrices appear in the article and need transcription, as Townsend and Landon's were for Section 11. The Rouder laboratory data is unpublished but public, with a master set, nested subsets and stimulus and response logged as global indices, which is the property that makes labels comparable across menus. Four large odor corpora are open and usable but carry a fixed four-descriptor menu, so they calibrate and cannot score.